ADR 1: Vector Database Selection

- **Status:** Accepted
- **Context:** FinServ Global requires an indexing engine capable of executing hybrid search (semantic dense embeddings + sparse keyword matching) across overlapping financial regulations. The system must support concurrent queries (10K/day) and enforce strict point-in-time isolation based on document version dates to support historical internal audits.
- **Decision:** Implement **Qdrant** self-hosted via Docker for development and deployed as a distributed StatefulSet on Kubernetes for production.
- **Alternatives Considered:** * *pgvector (PostgreSQL extension):* Rejected due to high relational locking overhead during large-scale concurrent indexing and lack of native out-of-the-box sparse-dense hybrid optimization.
    - *Pinecone (Managed Service):* Rejected because it is a closed-source, proprietary SaaS tool that violates data sovereignty guidelines by sending financial transaction lookups outside the corporate security perimeter.
- **Consequences:** Introduces a specialized vector database component into the infrastructure ecosystem. Requires the DevOps team to manage dedicated persistent volume claims (PVCs), snapshot backups, and monitoring metrics via Prometheus.